1. $A_q = \begin{pmatrix} 20 & -24 & -6 \\ 24 & -28 & -6 \\ -24 & 24 & 2 \end{pmatrix}$ $\quad \det|A_q - \lambda I| = 0$

$$\begin{vmatrix} 20-\lambda & -24 & -6 \\ 24 & -28-\lambda & -6 \\ -24 & 24 & 2-\lambda \end{vmatrix} = (20-\lambda)(-28-\lambda)(2-\lambda) - 24^2\cdot 6 - 24^2\cdot 6 + 24\cdot 6\cdot(28+\lambda) + 24^2(2-\lambda) + 6\cdot 24\cdot(20-\lambda) =$$

$$= (-560 + 8\lambda - 20\lambda + \lambda^2)(2-\lambda) - 24^2\cdot 12 + 4032 + 144\lambda + 1152 - 576\lambda + 2880 - 144\lambda =$$

$$= -1120 + 16\lambda + 2\lambda^2 + 560\lambda - 8\lambda^2 - \lambda^3 - 6912 + 4032 + 144\lambda + 1152 - 576\lambda + 2880 - 144\lambda =$$

$$= \lambda^3 + 6\lambda^2 - 32 = 0 \Rightarrow (\lambda-2)(\lambda+4)^2 = 0 \Rightarrow \sigma_q = \{-4; 2\}$$

$\lambda = 2, \lambda = -4 \longrightarrow$

$$\begin{pmatrix} 24 & -24 & -6 \\ 24 & -24 & -6 \\ -24 & 24 & 6 \end{pmatrix} \Rightarrow \xi^3 = 4\xi^1 - 4\xi^2 \Rightarrow \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} = C_1\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + C_2\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$$

$$\xi^1 = \tfrac{1}{4}\xi^3 + \xi^2$$

$$\begin{pmatrix} 18 & -24 & -6 \\ 24 & -30 & -6 \\ -24 & 24 & 0 \end{pmatrix} \sim \begin{pmatrix} 3 & -4 & -1 \\ 4 & -5 & -1 \\ -4 & 4 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & -1 \\ -1 & 1 & 0 \end{pmatrix} \Rightarrow$$

$$\Rightarrow \xi^2 = -\xi^3 \Rightarrow$$

$$\xi^1 = -\xi^3$$

$$\Rightarrow \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} = C_1\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

2. $A = \begin{pmatrix} 7 & 2 & 1 \\ -12 & -3 & -2 \\ -6 & -2 & 0 \end{pmatrix}$

$$\begin{vmatrix} 7-\lambda & 2 & 1 \\ -12 & -3-\lambda & -2 \\ -6 & -2 & -\lambda \end{vmatrix} = (7-\lambda)(3\lambda+\lambda^2) + 24 + 24 - 6(3+\lambda) - 24\lambda -$$

$$- 4(7-\lambda) = 21\lambda + 7\lambda^2 - 3\lambda^2 - \lambda^3 + 30 - 6\lambda - 24\lambda -$$

$$-28 + 4\lambda = \lambda^3 - 4\lambda^2 + 5\lambda - 2$$

$$\lambda = 1, \lambda = 2$$

$$\begin{pmatrix} 6 & 2 & 1 \\ -12 & -4 & -2 \\ -6 & -2 & -1 \end{pmatrix} \Rightarrow \begin{pmatrix} \xi^1 \\ \xi^2 \\ \xi^3 \end{pmatrix} = \begin{pmatrix} \xi^1 \\ \xi^2 \\ -6\xi^1 - 2\xi^2 \end{pmatrix} = C_1\begin{pmatrix} 1 \\ 0 \\ -6 \end{pmatrix} + C_2\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 2 & 1 \\ -12 & -5 & -2 \\ -6 & -2 & -2 \end{pmatrix} \sim \begin{pmatrix} -1 & 0 & -1 \\ -3 & -1 & -1 \\ 0 & -1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & -1 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} = C_1\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

3. $A = \begin{pmatrix} -78 & -72 & -24 & -48 \\ -6 & 6 & -12 & 12 \\ 114 & 96 & 42 & 60 \\ 78 & 60 & 36 & 30 \end{pmatrix} \Rightarrow \begin{vmatrix} -78-\lambda & -72 & -24 & -48 \\ -6 & 6-\lambda & -12 & 12 \\ 114 & 96 & 42-\lambda & 60 \\ 78 & 60 & 36 & 30-\lambda \end{vmatrix} \Rightarrow \sigma_q = \{6, -6\}$

$$\begin{pmatrix} -84 & -72 & -24 & -48 \\ -6 & 0 & -12 & 12 \\ 114 & 96 & 36 & 60 \\ 78 & 60 & 36 & 24 \end{pmatrix} \sim \begin{pmatrix} 7 & 6 & 2 & 4 \\ 1 & 0 & 2 & -2 \\ 19 & 16 & 6 & 10 \\ 13 & 10 & 6 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 & -2 \\ 0 & 6 & -12 & 18 \\ 0 & 16 & -32 & 48 \\ 0 & 10 & -20 & 30 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 & -2 \\ 0 & 1 & -2 & 3 \\ 0 & 1 & -2 & 3 \\ 0 & 1 & -2 & 3 \end{pmatrix} \Rightarrow$$

$\Rightarrow \begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{pmatrix} = C_1 \begin{pmatrix} -2 \\ 2 \\ 1 \\ 0 \end{pmatrix} + C_2 \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \end{pmatrix}$. Аналогично для $\lambda = -6$: $\begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{pmatrix} = C_1 \begin{pmatrix} -8 \\ 5 \\ 9 \\ 0 \end{pmatrix} + C_2 \begin{pmatrix} 2 \\ -8 \\ 0 \\ 9 \end{pmatrix}$

4. $A_q = \begin{pmatrix} 262 & 152 & 144 & 50 & 32 \\ -120 & -76 & -63 & -20 & -17 \\ -336 & -192 & -186 & -66 & -40 \\ -24 & -16 & -12 & -3 & -4 \\ -30 & 4 & -27 & -14 & 5 \end{pmatrix}$; $\in q = \{-2; 0; 1; 5\}$

$\lambda = -2$: $\begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \\ g_5 \end{pmatrix} = X_4 \begin{pmatrix} -19/48 \\ 17/32 \\ -105 \\ 1 \\ 0 \end{pmatrix} + X_5 \begin{pmatrix} 1/6 \\ -1/2 \\ 0 \\ 0 \\ 1 \end{pmatrix} = X_4 \begin{pmatrix} -38 \\ 51 \\ -10080 \\ 96 \\ 0 \end{pmatrix} + X_5 \begin{pmatrix} 1 \\ -3 \\ 0 \\ 0 \\ 6 \end{pmatrix}$

$\lambda = 0$: $\begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \\ g_5 \end{pmatrix} = X_5 \begin{pmatrix} -12 \\ 3 \\ 16 \\ 0 \\ 12 \end{pmatrix}$; $\lambda = 1$: $\begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \\ g_5 \end{pmatrix} = X_5 \begin{pmatrix} -2 \\ 1 \\ 2 \\ 1 \\ 1 \end{pmatrix}$; $\lambda = 5$: $\begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \\ g_5 \end{pmatrix} = X_5 \begin{pmatrix} 2 \\ -2 \\ -2 \\ -1 \\ 4 \end{pmatrix}$

5. $A_q = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 10 & -2 \\ 0 & -2 & 10 \end{pmatrix}$ $A_q^d = T^{-1} A_q T$, T - матрица перехода от ей $\Rightarrow$

$\begin{vmatrix} 4-\lambda & 0 & 0 \\ 0 & 10-\lambda & -2 \\ 0 & -2 & 10-\lambda \end{vmatrix} = (4-\lambda)(10-\lambda)(10-\lambda) - 4(4-\lambda) = (4-\lambda)(100 - 20\lambda + \lambda^2 - 4)$

$\lambda^2 - 20\lambda + 96$

$\lambda = 4; 8; 12$

$\Rightarrow (A - \lambda I)x = 0$

$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 6 & -2 \\ 0 & -2 & 6 \end{pmatrix} \Rightarrow \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix} = X_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $\left\{ \begin{pmatrix} -4 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix} = X_3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right.$

$\begin{pmatrix} -8 & 0 & 0 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{pmatrix} \Rightarrow \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix} = X_3 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$ $\left\{ T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix} \right.$

6. $A_q = \begin{pmatrix} 81 & -24 & -6 & -6 \\ 36 & -29 & -9 & -9 \\ -60 & 60 & 22 & 15 \\ 24 & -24 & -6 & 1 \end{pmatrix}$; $\in q = \{4; 7\}$

$\lambda = 4$: $\begin{cases} -60x_1 + 60x_2 + 18x_3 + 15x_4 = 0 \\ 15x_2 + 9x_3 = 0 \\ 6x_3 + 15x_4 = 0 \end{cases} \Rightarrow$ $x_3 = -\frac{5}{2}x_4$

$x_2 = -\frac{3}{5}x_3 = \frac{3}{2}x_4$

$60x_1 = 90x_4 - 45x_4 + 15x_4 \Rightarrow x_1 = x_4$

$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = C_4 \begin{pmatrix} 2 \\ 3 \\ -5 \\ 2 \end{pmatrix}$

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$\lambda = 7$: $24x_1 - 24x_2 - 6x_3 - 6x_4 = 0 \Rightarrow 4x_1 = 4x_2 + x_3 + x_4 \Rightarrow x_1 = x_2 + \frac{1}{4}x_3 + \frac{1}{4}x_4 \Rightarrow$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = C_2 \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + C_3 \begin{pmatrix} 1 \\ 0 \\ 4 \\ 0 \end{pmatrix} + C_4 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 4 \end{pmatrix}; \quad T = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 3 & 1 & 0 & 0 \\ -5 & 0 & 4 & 0 \\ 2 & 0 & 0 & 4 \end{pmatrix} \Rightarrow A_q^d = T^{-1} A_q T = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}$$

$$\cos(A_q^d) = \begin{pmatrix} \cos 4 & 0 & 0 & 0 \\ 0 & \cos 7 & 0 & 0 \\ 0 & 0 & \cos 7 & 0 \\ 0 & 0 & 0 & \cos 7 \end{pmatrix}; \quad \cos(q) = T \cos(A_q^d) T^{-1}$$